

XI UJIMQIN QI, China

WMO: 540120

Lat: 44.5703N

Lon: 117.6273E

Elev: 997

StdP: 89.91

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-28.1	-34.3	0.2	-30.0	-31.8	0.2	-27.4	13.0	-12.9	11.9	-11.7	1.9	270	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	31.4	17.0	29.3	16.6	27.6	16.1	20.0	26.4	19.0	25.5	18.1	24.4	3.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	14.6	23.1	16.9	13.6	21.6	15.9	12.8	21.1	62.3	26.3	58.6	25.3	55.5	24.4	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	9.2	8.0	-33.9	34.4	3.2	2.1	-36.1	35.9	-38.0	37.1	-39.8	38.3	-42.1	39.8
			-33.9	21.4	3.2	1.0	-36.2	22.1	-38.1	22.7	-39.9	23.3	-42.2	24.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-18.6	-14.3	-5.6	4.5	11.7	17.4	20.8	18.7	12.4	3.4	-7.4	-15.9
	DBStd	14.56	5.12	5.98	6.15	5.48	4.84	3.81	3.04	3.31	4.53	5.36	6.63	5.35
	HDD10.0	3833	886	682	484	179	36	1	0	0	27	213	523	803
	HDD18.3	6016	1144	915	742	414	213	61	9	34	185	463	773	1061
	CDD10.0	1036	0	0	1	15	89	223	333	269	97	8	0	0
	CDD18.3	177	0	0	0	1	8	33	84	45	6	0	0	0
	CDH23.3	1977	0	0	0	12	142	441	809	484	87	2	0	0
	CDH26.7	599	0	0	0	1	38	133	273	141	13	0	0	0
Wind	WSAvg	3.3	3.6	3.5	3.7	3.9	3.9	2.9	2.6	2.6	2.8	3.3	3.5	3.6
Precipitation	PrecAvg	333	2	3	6	10	27	50	98	80	33	16	5	3
	PrecMax	545	9	7	15	49	80	106	209	170	77	57	18	7
	PrecMin	189	0	0	0	0	4	9	25	21	5	0	0	0
	PrecStd	81	2	2	4	10	16	26	40	41	20	15	4	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.0	4.4	15.2	24.9	30.6	32.9	34.7	33.6	28.5	21.9	12.1	1.9
		MCWB	-4.9	-0.6	4.6	9.8	13.3	15.9	18.6	18.0	14.5	10.1	3.7	-1.8
	2%	DB	-5.2	1.5	10.6	20.9	27.2	30.0	31.8	30.1	25.9	18.6	8.2	-2.6
		MCWB	-7.5	-2.4	2.8	7.8	12.0	15.3	18.5	17.3	13.5	8.8	1.9	-5.1
	5%	DB	-7.6	-1.3	7.5	17.7	24.3	27.9	29.6	27.8	23.7	15.6	5.2	-5.2
		MCWB	-9.4	-4.4	0.8	6.4	10.9	15.1	18.0	16.6	12.7	7.0	0.2	-7.2
	10%	DB	-9.9	-4.1	4.5	15.0	21.7	25.9	27.7	25.9	21.4	12.9	2.6	-7.5
		MCWB	-11.4	-6.6	-0.8	5.2	10.0	14.5	17.6	16.1	12.0	5.5	-1.3	-9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.1	0.0	5.4	10.6	14.6	18.2	21.7	21.0	17.4	11.4	4.5	-1.9
		MCDB	-2.1	3.3	14.1	23.7	25.6	26.8	28.2	28.9	24.9	19.8	11.0	1.7
	2%	WB	-7.4	-2.3	3.0	8.5	13.1	17.3	20.6	19.5	15.2	9.3	2.0	-5.1
		MCDB	-5.4	1.5	10.2	19.0	24.0	24.8	27.0	25.9	21.9	17.0	7.6	-2.7
	5%	WB	-9.3	-4.4	1.0	7.0	12.0	16.5	19.6	18.5	14.0	7.7	0.4	-7.2
		MCDB	-7.6	-1.3	7.1	16.5	21.6	23.8	26.0	24.2	21.1	14.6	5.0	-5.3
	10%	WB	-11.3	-6.6	-0.8	5.6	10.7	15.7	18.7	17.6	12.8	6.0	-1.4	-9.2
		MCDB	-9.9	-4.2	4.3	14.5	20.2	23.0	25.3	23.0	19.8	12.3	2.5	-7.6
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.8	12.1	13.3	13.6	12.5	11.3	12.2	13.5	12.6	10.9	9.9
		MCDBR	13.3	14.6	16.2	17.8	18.0	16.1	14.5	15.3	16.4	16.7	13.8	12.4
		MCWBR	11.6	11.6	10.1	8.8	7.6	5.7	4.9	5.6	7.0	9.0	9.0	10.6
	5% WB	MCDBR	13.3	14.6	15.6	16.3	14.7	11.7	11.4	11.1	12.9	15.5	13.5	12.3
		MCWBR	11.7	11.7	9.9	8.7	6.7	4.8	4.7	4.9	6.2	8.8	9.1	10.5
Clear-Sky Solar Irradiance	taub	0.263	0.295	0.344	0.404	0.471	0.492	0.448	0.422	0.375	0.340	0.292	0.270	
	taud	2.294	2.220	2.138	2.030	1.897	1.951	2.146	2.184	2.243	2.249	2.276	2.258	
	Ebn at Noon	860	887	882	853	804	788	821	829	841	819	808	803	
	Edh at Noon	80	105	132	161	190	181	147	137	118	102	82	75	
All-Sky Solar Radiation	RadAvg	2.15	3.32	4.63	5.70	6.28	6.10	6.03	5.70	4.75	3.42	2.27	1.67	
	RadStd	0.17	0.20	0.28	0.37	0.41	0.50	0.31	0.20	0.28	0.28	0.17	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-1.99	-1.93	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	-1.59	N/A	N/A	-0.56	N/A	N/A	+68	N/A

Nomenclature: See separate page